

Effective Lasserre Exactness under a Regular-Sequence Homogenization Hypothesis

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Abstract

Hua and Qu obtained an explicit degree bound for exactness of Lasserre's hierarchy over a finite complex equality variety under a “no solutions at infinity” hypothesis. They asked whether an effective bound remains available when one assumes only that the homogenizations of the first n equality constraints form a regular sequence, even though the corresponding H -basis property may fail. We give a complete affirmative answer under the same finite-variety and nonsingularity hypotheses used by Hua and Qu.

Let $d_i = \deg g_i$ for $1 \leq i \leq n$, let

$$N := \prod_{i=1}^n d_i, \quad \nu := \sum_{i=1}^n (d_i - 1),$$

and let d_g and d_h be the maximum degrees of the extra equality constraints and the inequality constraints, respectively, with value 0 when the corresponding family is empty. If the equality Jacobian has rank n at every global minimizer, then the SOS relaxation is exact and attained at every order

$$d \geq \max \left\{ d_0, \left\lceil \frac{1}{2} \max \{ \deg f + N - 1, d_g + N - 1, d_h + N + \nu - 1 \} \right\rceil \right\},$$

where d_0 is the usual initial relaxation order. The proof replaces the missing H -basis property by a bounded saturation defect for the first n equations and, crucially, reduces all extra equality coefficients and SOS factors modulo those first n equations before applying the saturation bound. The result is a solution of the stated regular-sequence degree-bound problem, although the bound is not claimed to be sharp.

1 Introduction

Consider the polynomial optimization problem

$$f^* := \min \{ f(x) : g_1(x) = \cdots = g_k(x) = 0, \quad h_1(x) \geq 0, \dots, h_m(x) \geq 0 \}, \quad (1)$$

where $f, g_1, \dots, g_k, h_1, \dots, h_m \in \mathbb{R}[x_1, \dots, x_n]$, the feasible set is nonempty, and the complex equality variety is finite. Hua and Qu proved an explicit exactness order for Lasserre's hierarchy when the highest-degree parts of $g_1, \dots, g_n$ have no common nonzero complex zero. Their hypothesis implies two facts: the homogenizations of $g_1, \dots, g_n$ form a regular sequence, and $g_1, \dots, g_n$ form an H -basis. In the conclusion of their paper they asked for degree bounds assuming only the former property, because the latter can fail in the presence of solutions at infinity; see [2, Section 8].

The obstruction is entirely in degree-controlled ideal membership. If

$$I_0 := (g_1, \dots, g_n), \quad J := (g_1, \dots, g_k),$$

then the H -basis property would imply

$$p \in I_0, \quad \deg p = D \quad \implies \quad p = \sum_{i=1}^n \lambda_i g_i, \quad \deg(\lambda_i g_i) \leq D.$$

Under a regular-sequence hypothesis alone, one must generally allow an additive degree defect. We prove that the defect for the first n equations is the x_0 -saturation exponent of their homogenized complete intersection, and that this exponent is at most the scheme-theoretic multiplicity at infinity. Since the total projective degree is $N = \prod_i \deg g_i$, this gives a degree-only bound.

There is a second, indispensable point when $k > n$. One should not saturate the ideal generated by all displayed equality constraints. A theorem whose bound contains that all-generator saturation exponent is only partial progress, because the exponent is not controlled by the regular-sequence hypothesis and can depend badly on redundant generators. Instead, one first reduces the coefficients of $g_{n+1}, \dots, g_k$, as well as every polynomial appearing inside an SOS multiplier, modulo I_0 . The remaining residual belongs to I_0 , so only the uniformly controlled saturation of the complete intersection generated by the first n equations is needed.

Status of the result. The main theorem below is a complete solution of the degree-bound question posed by Hua and Qu, under the same finite-variety and rank hypotheses that they use to obtain an untruncated certificate. The resulting bound is intentionally conservative; no optimality claim is made.

2 Notation and the main result

Let

$$R := \mathbb{R}[x_1, \dots, x_n], \quad S := \mathbb{R}[x_0, x_1, \dots, x_n].$$

We use the convention $\deg 0 = -\infty$. For a nonzero $p \in R$ of degree a , its homogenization is

$$p^h(x_0, x) := x_0^a p(x/x_0) \in S_a.$$

Put

$$\bar{g}_i := g_i^h, \quad I_0 := (g_1, \dots, g_n) \subset R, \quad J := (g_1, \dots, g_k) \subset R,$$

and

$$K := (\bar{g}_1, \dots, \bar{g}_n) \subset S.$$

We assume throughout that $\bar{g}_1, \dots, \bar{g}_n$ form a regular sequence after extension of scalars to $\mathbb{C}$. Equivalently, $K_{\mathbb{C}} := K \mathbb{C}[x_0, \dots, x_n]$ is a homogeneous complete intersection of codimension n .

Set $h_0 := 1$. For $D \geq 0$, define

$$I(g_1, \dots, g_k)_D := \left\{ \sum_{i=1}^k \lambda_i g_i : \deg(\lambda_i g_i) \leq D \text{ for every } i \right\}$$

and

$$\text{QM}(h_0, \dots, h_m)_D := \left\{ \sum_{j=0}^m \sigma_j h_j : \sigma_j \in \sum R^2, \deg(\sigma_j h_j) \leq D \text{ for every } j \right\}.$$

The order- d SOS relaxation is

$$f_d^{\text{sos}} := \sup\{c \in \mathbb{R} : f - c \in I(g_1, \dots, g_k)_{2d} + \text{QM}(h_0, \dots, h_m)_{2d}\}.$$

Write

$$d_i := \deg g_i \quad (1 \leq i \leq n), \quad N := \prod_{i=1}^n d_i, \quad \nu := \sum_{i=1}^n (d_i - 1). \quad (2)$$

Also set

$$d_g := \max(\{0\} \cup \{\deg g_i : n < i \leq k\}), \quad d_h := \max(\{0\} \cup \{\deg h_j : 1 \leq j \leq m\}), \quad (3)$$

and let

$$d_0 := \left\lceil \frac{1}{2} \max(\deg f, \max_i \deg g_i, \max_j \deg h_j) \right\rceil, \quad (4)$$

where an empty maximum is omitted.

Theorem 2.1 (Degree-only exactness bound). *Assume that $V_{\mathbb{C}}(J)$ is finite, the feasible set of (1) is nonempty, and $\bar{g}_1, \dots, \bar{g}_n$ form a regular sequence. Suppose that for every global minimizer u of (1),*

$$\text{rank}[\nabla g_1(u) \ \cdots \ \nabla g_k(u)] = n. \quad (5)$$

Then the SOS relaxation is exact and its optimum is attained at every order

$$d \geq \max \left\{ d_0, \left\lceil \frac{B_{\text{deg}}}{2} \right\rceil \right\}, \quad B_{\text{deg}} := \max\{\deg f + N - 1, d_g + N - 1, d_h + N + \nu - 1\}. \quad (6)$$

At the same orders, the moment relaxation is also exact and attained.

The rest of the paper proves Theorem 2.1. We first establish a sharper bound involving two computable invariants of I_0 ; the degree-only formula (6) is then an immediate consequence.

3 Interpolation and saturation for the first n equations

Let

$$A_0 := R/I_0, \quad \ell := \dim_{\mathbb{R}} A_0.$$

The regular-sequence hypothesis implies that A_0 is finite-dimensional. We use the following two invariants.

Definition 3.1 (Interpolation degree). *The interpolation degree of I_0 is*

$$\mu := \min\{t \geq 0 : R_{\leq t} \longrightarrow A_0 \text{ is surjective}\}.$$

Thus every residue class modulo I_0 has a representative of degree at most μ .

Definition 3.2 (Saturation exponent). *Let $S_{\mathbb{C}} := \mathbb{C}[x_0, \dots, x_n]$. The x_0 -saturation exponent of $K_{\mathbb{C}}$ is*

$$\tau := \min\{s \geq 0 : x_0^s(K_{\mathbb{C}} : x_0^\infty) \subseteq K_{\mathbb{C}}\}. \quad (7)$$

It is finite because $(K_{\mathbb{C}} : x_0^\infty)/K_{\mathbb{C}}$ is a finitely generated x_0 -torsion module.

The next proposition is the main projective input.

Proposition 3.3 (Uniform bounds on μ and τ). *Under the regular-sequence hypothesis,*

$$\mu \leq \min\{\nu, \ell - 1\}, \quad \tau \leq N - \ell. \quad (8)$$

Consequently,

$$\tau \leq N - 1, \quad \mu + \tau \leq N - 1, \quad 2\mu + \tau \leq N + \nu - 1. \quad (9)$$

Proof. We divide the proof into three steps.

Step 1: the projective complete intersection. Let

$$B := S_{\mathbb{C}}/K_{\mathbb{C}}, \quad X := \text{Proj } B.$$

Since $K_{\mathbb{C}}$ is generated by a homogeneous regular sequence of degrees $d_1, \dots, d_n$, the Koszul resolution gives

$$H_B(z) = \frac{\prod_{i=1}^n (1 - z^{d_i})}{(1 - z)^{n+1}} = \frac{\prod_{i=1}^n (1 + z + \dots + z^{d_i-1})}{1 - z}. \quad (10)$$

Hence X is a zero-dimensional projective complete intersection of degree N , and

$$\text{reg } B = \nu. \quad (11)$$

Moreover, B is Cohen–Macaulay of dimension one, so $K_{\mathbb{C}}$ is saturated with respect to the irrelevant ideal.

Let

$$U := X \cap D_+(x_0).$$

The dehomogenization isomorphism gives

$$\Gamma(U, \mathcal{O}_U) \cong (B_{x_0})_0 \cong \mathbb{C}[x_1, \dots, x_n]/(I_0)_{\mathbb{C}}. \quad (12)$$

Thus U has length ℓ . Since X is zero-dimensional, U is both open and closed. Let $Z := X \setminus U$ with the complementary open-and-closed scheme structure. Then

$$\text{length}(Z) = N - \ell. \quad (13)$$

Because the feasible set is nonempty, U is nonempty and therefore $1 \leq \ell \leq N$.

Step 2: interpolation. For every $t \geq \nu$, Castelnuovo–Mumford regularity gives a surjection

$$B_t \longrightarrow H^0(X, \mathcal{O}_X(t)). \quad (14)$$

In fact it is an isomorphism, but surjectivity is all that is needed. Let $a \in \Gamma(U, \mathcal{O}_U)$. Since x_0 is invertible on U , it trivializes $\mathcal{O}_X(t)|_U$, so a determines a section of $\mathcal{O}_X(t)|_U$. Because U is open and closed, this section extends by zero on Z to a section of $\mathcal{O}_X(t)$. Taking $t = \nu$ in (14), we obtain a homogeneous form $F \in S_{\mathbb{C}, \nu}$ whose restriction to U , divided by x_0^ν , equals a . Equivalently, $F(1, x)$ represents a modulo $(I_0)_{\mathbb{C}}$ and has degree at most ν . Thus the complexification of the map $R_{\leq \nu} \rightarrow A_0$ is surjective. Faithful flatness of $\mathbb{C}/\mathbb{R}$ implies that the original real map is surjective, and hence

$$\mu \leq \nu. \quad (15)$$

A second elementary bound uses standard monomials. Choose any monomial order for $(I_0)_{\mathbb{C}}$, and let $\mathcal{B}$ be its set of standard monomials. Then $\mathcal{B}$ is divisor-closed and $|\mathcal{B}| = \ell$. If $x^\alpha \in \mathcal{B}$ has

total degree q , repeatedly deleting one variable gives a chain of $q + 1$ distinct divisors, all lying in $\mathcal{B}$. Hence $q + 1 \leq \ell$. Every residue class is a linear combination of the standard monomials, so

$$\mu \leq \ell - 1. \quad (16)$$

Together, (15) and (16) give the first inequality in (8).

Step 3: the saturation exponent. Choose a linear form $L \in S_{\mathbb{C},1}$ that is nonzero at every closed point of X ; this is possible because X has finite support. Since B is Cohen–Macaulay of dimension one and L lies in no minimal prime of B , multiplication by L is injective on B . Moreover $X \subset D_+(L)$, and therefore

$$C := (B_L)_0 = \Gamma(X, \mathcal{O}_X) \quad (17)$$

is an Artinian $\mathbb{C}$ -algebra of length N .

Put

$$z := x_0/L \in C.$$

The open-and-closed decomposition $X = U \sqcup Z$ induces an algebra decomposition

$$C \cong C_{\text{aff}} \times C_{\infty}, \quad \text{length}(C_{\text{aff}}) = \ell, \quad \text{length}(C_{\infty}) = N - \ell, \quad (18)$$

where z is a unit on C_{aff} and is nilpotent on C_{∞} .

We claim that τ is precisely the nilpotence index of z on C_{∞} . Let $F \in S_{\mathbb{C}}$ be homogeneous of degree d , and write

$$f_F := F/L^d \in C.$$

Because L is a nonzerodivisor on B , for every $q \geq 0$ we have

$$x_0^q F \in K_{\mathbb{C}} \iff z^q f_F = 0 \text{ in } C. \quad (19)$$

Indeed, the forward implication is immediate. Conversely, if $z^q f_F = 0$, then some power of L annihilates the class of $x_0^q F$ in B ; injectivity of multiplication by L forces that class itself to be zero. It follows from (19) that

$$F \in K_{\mathbb{C}} : x_0^{\infty} \iff f_F \in 0 \times C_{\infty}.$$

Every element of C_{∞} is represented by some homogeneous fraction F/L^d . Consequently, the least s satisfying $x_0^s(K_{\mathbb{C}} : x_0^{\infty}) \subseteq K_{\mathbb{C}}$ is the least s satisfying $z^s C_{\infty} = 0$.

Let this nilpotence index be e . Since z belongs to the Jacobson radical of the Artinian ring C_{∞} , the chain

$$C_{\infty} \supset zC_{\infty} \supset \cdots \supset z^{e-1}C_{\infty} \supset 0$$

is strict. Indeed, an equality $z^i C_{\infty} = z^{i+1} C_{\infty}$ with $i < e$ would give $z^i C_{\infty} = z(z^i C_{\infty})$, and Nakayama's lemma would force $z^i C_{\infty} = 0$, contrary to the definition of e . Hence

$$\tau = e \leq \text{length}(C_{\infty}) = N - \ell.$$

This proves (8).

Finally,

$$\tau \leq N - \ell \leq N - 1,$$

$$\mu + \tau \leq (\ell - 1) + (N - \ell) = N - 1,$$

and

$$2\mu + \tau = \mu + (\mu + \tau) \leq \nu + N - 1.$$

Thus (9) follows. $\square$

Remark 3.4. Both μ and τ are computable. A degree-compatible Gröbner basis of I_0 gives μ from the largest degree of a standard monomial. The exponent τ is the least s for which multiplication by x_0^s annihilates $(K_{\mathbb{C}} : x_0^{\infty})/K_{\mathbb{C}}$.

4 A bounded-defect replacement for the H -basis property

The following lemma is the precise substitute for degree-compatible ideal membership.

Lemma 4.1 (Saturation-controlled membership). *Let $p \in I_0$ and let $D := \deg p$. Then there exist $\lambda_1, \dots, \lambda_n \in R$ such that*

$$p = \sum_{i=1}^n \lambda_i g_i \quad (20)$$

and

$$\deg(\lambda_i g_i) \leq D + \tau \quad (1 \leq i \leq n). \quad (21)$$

In particular, one may replace τ by the degree-only bound $N - \ell \leq N - 1$.

Proof. Choose any representation $p = \sum_{i=1}^n a_i g_i$. Homogenizing this identity to a sufficiently large common degree gives an integer $q \geq 0$ such that

$$x_0^q p^h \in K.$$

Hence $p^h \in K_{\mathbb{C}} : x_0^{\infty}$. By the definition of τ ,

$$x_0^{\tau} p^h \in K_{\mathbb{C}}.$$

The left-hand side has real coefficients. Since extension from $\mathbb{R}$ to $\mathbb{C}$ is faithfully flat,

$$K_{\mathbb{C}} \cap S = K,$$

so $x_0^{\tau} p^h \in K$. It is homogeneous of degree $D + \tau$. Because K is generated by the homogeneous forms $\bar{g}_i$, its degree- $(D + \tau)$ component is

$$K_{D+\tau} = \sum_{i=1}^n S_{D+\tau-d_i} \bar{g}_i.$$

Thus there are homogeneous $\Lambda_i \in S$ such that

$$x_0^{\tau} p^h = \sum_{i=1}^n \Lambda_i \bar{g}_i, \quad \deg(\Lambda_i \bar{g}_i) = D + \tau$$

for every nonzero term. Dehomogenizing at $x_0 = 1$ gives (20), with $\lambda_i(x) = \Lambda_i(1, x)$, and yields (21). $\square$

Remark 4.2 (Exact meaning of the saturation exponent). *The integer τ is the least uniform additive defect for the displayed generators $g_1, \dots, g_n$. Lemma 4.1 proves one direction. Conversely, suppose that an integer s has the property that every $p \in I_0$ admits a representation*

$$p = \sum_{i=1}^n \lambda_i g_i, \quad \deg(\lambda_i g_i) \leq \deg p + s.$$

Let $F \in K_{\mathbb{C}} : x_0^{\infty}$ be nonzero and homogeneous of degree d , and put $p := F(1, x)$. Then $p \in (I_0)_{\mathbb{C}}$. By complexifying the assumed real degree-defect property, the same bound holds for elements of $(I_0)_{\mathbb{C}}$. If $e := \deg p$, homogeneity gives $F = x_0^{d-e} p^h$. Homogenizing a degree-defect- s representation of p to degree $e + s$ yields $x_0^s p^h \in K_{\mathbb{C}}$, and hence $x_0^s F \in K_{\mathbb{C}}$. Thus $s \geq \tau$.

5 Truncating arbitrary certificates

We now show how to control a certificate involving all equality and inequality constraints. This is the step that makes the argument work for arbitrary $k \geq n$.

Theorem 5.1 (Effective truncation under a regular sequence). *Let $p \in J + \text{QM}(h_0, \dots, h_m)$. Define*

$$B_{\text{ref}}(p) := \max\{\deg p + \tau, \mu + \tau + d_g, 2\mu + \tau + d_h\}. \quad (22)$$

Then

$$p \in I(g_1, \dots, g_k)_{B_{\text{ref}}(p)} + \text{QM}(h_0, \dots, h_m)_{B_{\text{ref}}(p)}. \quad (23)$$

Consequently,

$$B_{\text{ref}}(p) \leq B_{\text{deg}}(p) := \max\{\deg p + N - 1, d_g + N - 1, d_h + N + \nu - 1\}. \quad (24)$$

Proof. Choose an untruncated representation

$$p = \sum_{i=1}^k a_i g_i + \sum_{j=0}^m \sigma_j h_j, \quad \sigma_j = \sum_{t=1}^{r_j} b_{jt}^2. \quad (25)$$

For each $i > n$, choose a representative $\tilde{a}_i \in R_{\leq \mu}$ of the residue class of a_i modulo I_0 . For each j, t , choose $\tilde{b}_{jt} \in R_{\leq \mu}$ representing b_{jt} modulo I_0 , and set

$$\tilde{\sigma}_j := \sum_{t=1}^{r_j} \tilde{b}_{jt}^2.$$

Then

$$a_i - \tilde{a}_i \in I_0 \quad (i > n),$$

and

$$b_{jt}^2 - \tilde{b}_{jt}^2 = (b_{jt} - \tilde{b}_{jt})(b_{jt} + \tilde{b}_{jt}) \in I_0.$$

Consequently,

$$\sigma_j - \tilde{\sigma}_j \in I_0.$$

Therefore

$$\phi := p - \sum_{i=n+1}^k \tilde{a}_i g_i - \sum_{j=0}^m \tilde{\sigma}_j h_j \in I_0. \quad (26)$$

Moreover,

$$\deg(\tilde{a}_i g_i) \leq \mu + d_g, \quad \deg(\tilde{\sigma}_j h_j) \leq 2\mu + d_h.$$

Hence

$$\deg \phi \leq D := \max\{\deg p, \mu + d_g, 2\mu + d_h\}. \quad (27)$$

Applying Lemma 4.1 to ϕ gives

$$\phi = \sum_{i=1}^n \lambda_i g_i, \quad \deg(\lambda_i g_i) \leq D + \tau.$$

Substituting this identity into (26) proves (23), because

$$D + \tau = B_{\text{ref}}(p).$$

Finally, Proposition 3.3 gives

$$\deg p + \tau \leq \deg p + N - 1,$$

$$\mu + \tau + d_g \leq N - 1 + d_g,$$

and

$$2\mu + \tau + d_h \leq N + \nu - 1 + d_h.$$

Taking the maximum yields (24). $\square$

Remark 5.2 (Why the extra generators cause no uncontrolled saturation). *A saturation exponent attached to the full displayed generating system $g_1, \dots, g_k$ can depend badly on redundant high-degree generators. Theorem 5.1 avoids this issue: coefficients of the extra generators are first reduced modulo I_0 , and saturation is invoked only for the complete intersection generated by $g_1, \dots, g_n$. Its exponent is bounded by $N - \ell$.*

6 Proof of Lasserre exactness

The positivity input is exactly the one already established by Hua and Qu.

Proposition 6.1 (Hua–Qu untruncated certificate). *Assume that $V_{\mathbb{C}}(J)$ is finite and that (5) holds at every global minimizer. Then there exist real polynomials $q_1, \dots, q_{m+2}$ and $\lambda_1, \dots, \lambda_k$ such that*

$$f - f^* = q_1^2 + q_2^2 + \sum_{i=1}^k \lambda_i g_i + \sum_{j=1}^m q_{j+2}^2 h_j. \quad (28)$$

In particular,

$$f - f^* \in J + \text{QM}(h_0, \dots, h_m).$$

Proof. This is [2, Corollary 1]. Its proof uses only the finiteness of the complex equality variety and the rank condition at the global minimizers; the no-solutions-at-infinity hypothesis enters later, when Hua and Qu impose degree control. $\square$

Proof of Theorem 2.1. Apply Theorem 5.1 to $p = f - f^*$. Since $\deg(f - f^*) \leq \deg f$, Proposition 6.1 and (24) give

$$f - f^* \in I(g_1, \dots, g_k)_{B_{\deg}} + \text{QM}(h_0, \dots, h_m)_{B_{\deg}},$$

where $B_{\deg}$ is defined in (6). Thus, whenever $2d \geq B_{\deg}$, the value $c = f^*$ is feasible in the order- d SOS relaxation. Conversely, evaluating any feasible identity $f - c \in I(g)_{2d} + \text{QM}(h)_{2d}$ at a feasible point gives $c \leq f$ at that point and hence $c \leq f^*$. Therefore

$$f_d^{\text{SOS}} = f^*,$$

and the supremum is attained by $c = f^*$.

For the moment relaxation, evaluation at any global minimizer is a feasible moment functional with objective value f^* . Weak duality gives

$$f_d^{\text{SOS}} \leq f_d^{\text{mom}} \leq f^*.$$

Hence $f_d^{\text{mom}} = f^*$, attained by that evaluation functional. Including $d \geq d_0$ gives the standard hierarchy convention and completes the proof. $\square$

7 Comparison with the no-infinity case

The refined bound (22) clarifies exactly what changes when points at infinity are allowed.

Corollary 7.1 (Refined exactness order). *Under the hypotheses of Theorem 2.1, exactness and attainment already hold for every*

$$d \geq \max \left\{ d_0, \left\lceil \frac{1}{2} \max \{ \deg f + \tau, \mu + \tau + d_g, 2\mu + \tau + d_h \} \right\rceil \right\}.$$

Proof. Use Theorem 5.1 directly, without replacing μ and τ by their universal bounds. □

If the projective complete intersection has no component at infinity, then $\ell = N$ and $\tau = 0$. Proposition 3.3 gives $\mu \leq \nu$, so Corollary 7.1 reduces to

$$2d \geq \max \{ \deg f, \nu + d_g, 2\nu + d_h \},$$

which is the degree estimate underlying Hua and Qu’s theorem. Thus the regular-sequence argument genuinely extends their method: the only new cost is the saturation defect τ , while the coefficient-reduction step remains unchanged.

Remark 7.2 (Coarseness and possible refinements). *The degree-only bound in Theorem 2.1 uses only*

$$\mu \leq \min \{ \nu, \ell - 1 \}, \quad \tau \leq N - \ell.$$

In concrete systems, μ and τ can be substantially smaller. Computing them gives the sharper bound of Corollary 7.1. The present argument therefore resolves existence of an explicit bound, while leaving open the separate problem of finding the best possible dependence on the input degrees and the geometry at infinity.

8 Conclusion

The regular-sequence hypothesis alone is sufficient for an explicit Lasserre exactness order. The proof has two algebraic components:

- (i) the first n homogenized equations define a projective complete intersection of degree N , whose affine interpolation degree is at most ν and whose x_0 -saturation exponent is at most its multiplicity at infinity;
- (ii) every extra equality coefficient and every polynomial inside an SOS multiplier can be reduced modulo the first n equations before applying the saturation-controlled membership bound.

Together with the untruncated Hua–Qu certificate, these facts yield the explicit order (6). Accordingly, the regular-sequence degree-bound question is solved under the original finite-variety and rank assumptions.

References

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